

EE 156 – CHAPTER 1

Fundamentals of Electrical Controls and PLC Systems

Introduction to Electrical Control Systems

1.1 DEFINITION AND PURPOSE

What is an Electrical Control System?

An **electrical control system** is a combination of electrical devices used to:

- **Regulate** – maintain desired operating conditions
- **Direct** – control the sequence of operations
- **Manage** – oversee the operation of electrical equipment or industrial processes

Purpose of Electrical Control Systems

Function	Description	Example
Operate	Start/stop machines	Motor starter
Regulate	Maintain parameters	Speed controller
Monitor	Check conditions	Temperature sensor
Protect	Prevent damage	Circuit breaker, overload relay

Objectives of Electrical Control Systems

- 1 **Control equipment operation** – Start, stop, accelerate, decelerate
- 2 **Maintain process variables** – Temperature, pressure, speed, flow
- 3 **Ensure safety** – Protect personnel and equipment
- 4 **Improve efficiency** – Optimize energy usage
- 5 **Enhance productivity** – Automate repetitive tasks
- 6 **Provide monitoring** – Display system status

1.2 APPLICATIONS OF ELECTRICAL CONTROL SYSTEMS

Industrial Applications

Application	Example
Manufacturing	Assembly line control, CNC machines
Material handling	Conveyor systems, cranes, hoists

Pumping	Water treatment plants, irrigation
HVAC	Heating, ventilation, air conditioning
Processing	Chemical plants, food processing
Packaging	Bottling lines, labeling machines

Residential Applications

- **Lighting control** – Automatic lighting, dimmers
- **HVAC** – Thermostats, air conditioning
- **Appliance control** – Washing machines, dishwashers, ovens
- **Security systems** – Alarms, access control
- **Entertainment** – Audio/video systems

Commercial Applications

- **Building automation** – Smart buildings
- **Elevators/escalators** – Passenger transport
- **Fire detection** – Fire alarms, sprinkler systems
- **Access control** – Card readers, biometric systems
- **Lighting** – Office and street lighting

Power System Applications

- **Power generation** – Turbine control, excitation systems
- **Power transmission** – Substation automation
- **Power distribution** – Switchgear, protection relays
- **Renewable energy** – Solar tracking, wind turbine control
- **Smart grids** – Demand response, load management

1.3 CLASSIFICATION OF CONTROL SYSTEMS

Overview

Control systems are broadly classified into:

- 1 **Manual vs Automatic** (based on human involvement)
- 2 **Open-loop vs Closed-loop** (based on feedback)

1.3.1 Manual Control Systems

Definition: A control system where a **human operator directly controls** the operation of the machine or process.

Characteristics:

- Human intervention is required
- Operator makes decisions and takes actions
- No automatic feedback or correction

Examples:

Example	Description
Manual light switching	Operator turns switch ON/OFF
Hand-operated water pump	Operator pumps manually
Hand-operated drilling machine	Operator controls speed/feed
Manual gear shift in car	Driver selects gears

Advantages:

- 1 **Simple** – Easy to understand and implement
- 2 **Low cost** – No complex automation equipment needed
- 3 **Easy to operate** – Minimal training required

Disadvantages:

- 1 **Constant human involvement** – Cannot work unattended
- 2 **Less accurate** – Subject to human error
- 3 **Slower operation** – Limited by human speed
- 4 **Higher chance of human error** – Fatigue, distraction, mistakes

1.3.2 Automatic Control Systems

Definition: A control system where machines operate **automatically with little or no human intervention**.

Characteristics:

- No continuous human involvement
- Uses sensors and controllers
- Makes decisions based on setpoints

Examples:

Example	Description
Automatic washing machine	Cycles through wash, rinse, spin automatically
Automatic voltage regulator (AVR)	Maintains constant voltage
Automatic streetlights	Turn on/off based on light level
PLC-controlled industrial system	Fully automated production

Advantages:

- 1 **High efficiency** – Optimizes operations
- 2 **Faster operation** – Can operate at high speeds
- 3 **Improved accuracy** – Precise control
- 4 **Reduced human error** – Consistent performance
- 5 **Increased safety** – Remote/hazardous operation

Disadvantages:

- 1 **Higher cost** – More expensive to install
- 2 **More complex** – Requires sophisticated design
- 3 **Requires technical expertise** – Skilled maintenance needed

1.4 OPEN-LOOP CONTROL SYSTEMS

Definition

A control system where the **output does not affect the control action**. There is **no feedback** from the output.

Key Feature

- Control action is independent of output
- Cannot detect or correct errors
- Also called **non-feedback control**

Block Diagram of Open-Loop System

Input → [Controller] → [Actuator] → [Process] → Output
↑
(No feedback path)

Examples of Open-Loop Systems

Application	Input	Output	Why Open-Loop
Electric kettle	ON/OFF switch	Water heated	No temperature feedback
Traffic light	Timer setting	Light sequence	Fixed timing, no traffic sensing
Bread toaster	Timer setting	Toast done	Fixed time, no color sensing
Washing machine timer	Time setting	Wash cycle	Fixed cycle, no cleanliness feedback
Conveyor belt	Speed setting	Belt moves	Fixed speed, no load feedback
Heating element	Voltage setting	Heat output	No temperature measurement
Industrial oven (simple)	Temperature setpoint	Heat generated	No actual temperature feedback

Advantages:

- 1 **Simple** – Easy to design and build
- 2 **Low-cost** – Less expensive components
- 3 **Easy maintenance** – Fewer components to fail

Disadvantages:

- 1 **Lower accuracy** – Cannot maintain precise output
- 2 **Cannot automatically correct errors** – No self-correction
- 3 **Affected by disturbances** – No way to compensate

4 **Requires recalibration** – Periodic adjustments needed

1.5 CLOSED-LOOP CONTROL SYSTEMS

Definition

A control system where the **output is measured and compared** with the desired input (setpoint). The difference (error) is used to adjust the control action.

Key Feature

- Uses **feedback** from the output
- Can detect and correct errors
- Also called **feedback control system**

Block Diagram of Closed-Loop System

```
Setpoint → [Σ] → [Controller] → [Actuator] → [Process] → Output
↑ |
| |
←----- [Feedback/Sensor] -----→
```

Examples of Closed-Loop Systems

Application	Controlled Variable	Sensor	How It Works
Air conditioner	Room temperature	Thermistor	Maintains set temperature
Automatic voltage regulator	Output voltage	Voltage sensor	Keeps voltage constant
Cruise control	Vehicle speed	Speed sensor	Maintains set speed
Automatic water heater	Water temperature	Thermostat	Maintains set temperature
Room thermostat	Room temperature	Bimetallic strip	Turns heater on/off
Motor speed control	Motor speed	Tachometer	Maintains set speed
Process control	Product quality	Sensor	Maintains quality
Robot arm	Position	Encoder	Follows programmed path
Elevator control	Floor position	Position sensor	Stops at correct floor

Advantages:

- 1 **High accuracy** – Precise control
- 2 **Self-correcting** – Compensates for disturbances
- 3 **Better performance** – Stable and reliable
- 4 **Adaptive** – Can handle varying conditions

Disadvantages:

- 1 **More expensive** – Additional components needed
- 2 **More complex** – Harder to design and implement
- 3 **Potential instability** – Can oscillate if not designed properly

1.6 COMPARISON: OPEN-LOOP vs CLOSED-LOOP

Feature	Open-Loop System	Closed-Loop System
Feedback	No feedback	Uses feedback
Accuracy	Lower accuracy	High accuracy
Error correction	Cannot correct errors	Self-correcting
Disturbance handling	Affected by disturbances	Compensates for disturbances
Stability	Always stable	Can be unstable
Cost	Low cost	Higher cost
Complexity	Simple	Complex
Maintenance	Easy	Requires expertise
Human involvement	May need periodic adjustment	Minimal after setup
Example	Toaster, traffic light	Air conditioner, AVR

1.7 COMPONENTS OF ELECTRICAL CONTROL SYSTEMS

Major Components

Component	Function	Examples
Input Devices	Sense conditions	Pushbuttons, switches, sensors
Controllers	Process decisions	Relays, PLCs, timers, counters
Output Devices	Execute actions	Contactors, relays, solenoids, motors
Power Supply	Provide power	Batteries, power supplies, transformers
Human-Machine Interface (HMI)	Display/control	Screens, touch panels, indicators
Feedback Devices	Measure outputs	Thermocouples, encoders, sensors

Detailed Component Breakdown

Input Devices

Device	Function	Application
Pushbutton	Manual switch	Start/stop control
Limit switch	Position detection	Machine travel limits
Proximity sensor	Object presence	Conveyor objects
Photoelectric sensor	Light beam detection	Safety guards
Temperature sensor	Temperature measurement	HVAC, process control
Pressure switch	Pressure detection	Pump control
Float switch	Liquid level	Tank filling
Encoder	Position/speed	Motor feedback

Controllers

Device	Function	Application
Relay	Electromagnetic switching	Logic control
Timer	Time delay control	Sequencing
Counter	Count events	Production monitoring
PLC	Programmable control	Industrial automation

Microcontroller	Embedded control	Consumer electronics
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Output Devices

Device	Function	Application
Contactor	High-power switching	Motor starting
Relay	Control switching	Signal control
Solenoid	Linear actuation	Valve control
Motor	Rotation	Motion control
Indicating lamp	Visual signal	Status display
Alarm	Audio signal	Warning system
Heater	Heat generation	Process heating

1.8 OPERATION OF ELECTRICAL CONTROL SYSTEMS

Basic Operation Flow

- 1 **Input/Detection** → Sensor or switch detects condition
- 2 **Decision/Processing** → Controller compares to setpoint
- 3 **Action/Output** → Actuator executes required action
- 4 **Feedback** → Sensor measures result (in closed-loop)

Example: Automatic Street Light Control

Photocell detects darkness → Controller decides → Light turns ON
↓
(No feedback needed - open-loop system)

Example: Room Temperature Control

Thermostat measures temperature → Compare to setpoint → If too cold, turn heater ON
↑ ↓
←----- Temp feedback (closed-loop) -----→

Example: Conveyor Belt Control with PLC

Start button pressed → PLC receives signal → Program runs
↓
Motor contactor energized
↓
Conveyor belt moves
↓
Products transported

1.9 SAFETY IN ELECTRICAL CONTROL SYSTEMS

Importance of Safety

- 1 **Protect personnel** from electric shock and injury
- 2 **Protect equipment** from damage
- 3 **Prevent fire hazards**
- 4 **Ensure reliable operation**
- 5 **Comply with regulations**

Safety Precautions

Precaution	Description	Purpose
Proper grounding	Earth all metal parts	Prevent electric shock
Overcurrent protection	Fuses, circuit breakers	Prevent overheating and fire

Emergency stop	E-stop pushbuttons	Immediate shutdown
Interlocking	Prevent conflicting operations	Avoid dangerous conditions
Lockout/Tagout	Lock power during maintenance	Prevent accidental startup
Use of safety relays	Safety-rated components	Ensure reliable protection
Barriers/guards	Physical protection	Prevent contact with hazards
Warning labels	Clear signage	Inform of dangers
Personal Protective Equipment (PPE)	Safety gear	Protect workers

Safety Standards and Symbols

Common Standards

Standard	Organization	Focus
IEC	International Electrotechnical Commission	International standards
IEEE	Institute of Electrical and Electronics Engineers	Electrical/electronic standards
NEMA	National Electrical Manufacturers Association	US equipment standards
OSHA	Occupational Safety and Health Administration	Workplace safety
NFPA	National Fire Protection Association	Fire safety

Why Standards are Important

- 1 **Uniformity** – Consistent symbols and diagrams
- 2 **Safety** – Standardized safety requirements
- 3 **Easy troubleshooting** – Universal understanding
- 4 **Proper communication** – Common language for engineers

1.10 LIMITATIONS OF ELECTRICAL CONTROL SYSTEMS

Limitation	Description
Initial installation cost	High upfront investment
Complexity	Requires expert design
Maintenance requirements	Regular servicing needed
Need for skilled personnel	Trained operators/technicians required
Possible system failures	Can cause downtime
Power dependency	Require reliable power supply
EMI/EMC issues	Susceptible to electromagnetic interference

1.11 EMERGING TRENDS IN ELECTRICAL CONTROL SYSTEMS

Industry 4.0 Technologies

Trend	Description	Application
Smart automation	Intelligent self-optimizing systems	Smart factories
Industrial IoT	Connected devices and sensors	Remote monitoring

Artificial Intelligence	Machine learning for control	Predictive maintenance
Wireless monitoring	Wireless sensors and networks	Data collection
Smart sensors	Intelligent sensing devices	Condition monitoring
Remote control	Control from anywhere	Teleoperation
Cloud computing	Data storage and analysis	Big data analytics
Digital twins	Virtual replicas	Simulation and testing

1.12 REVIEW QUESTIONS AND ANSWERS

1. Define an electrical control system.

Answer: An electrical control system is a combination of electrical devices used to regulate, direct, or manage the operation of electrical equipment or industrial processes.

Key components: Input devices, controllers, output devices, power supply, and feedback elements.

2. State four applications of electrical control systems.

Answer:

1. **Industrial** – Assembly line control, CNC machines
2. **Residential** – Lighting control, HVAC systems, appliances
3. **Commercial** – Building automation, elevators, fire detection
4. **Power systems** – Substation automation, generator control

3. Differentiate between manual and automatic control systems.

Aspect	Manual Control	Automatic Control
Human involvement	Constant	Little or none
Decision making	Human operator	Controller
Accuracy	Lower	Higher
Speed	Slower	Faster
Cost	Low	High
Example	Manual switch	PLC system

4. Define an open-loop control system.

Answer: A control system where the **output does not affect the control action**. There is no feedback from the output.

Example: Bread toaster – fixed timing regardless of whether toast is done.

Characteristics: No feedback; cannot correct errors; simple and inexpensive.

5. Define a closed-loop control system.

Answer: A control system where the **output is measured and compared** with the desired input. The difference (error) is used to adjust the control action.

Example: Air conditioner – measures room temperature and adjusts cooling.

Characteristics: Uses feedback; self-correcting; high accuracy.

6. List four components of an electrical control system.

Answer:

1. **Input devices** – Pushbuttons, switches, sensors
2. **Controllers** – Relays, PLCs, timers
3. **Output devices** – Contactors, motors, solenoids
4. **Power supply** – Batteries, power supplies

7. State four safety precautions in electrical control systems.

Answer:

1. **Proper grounding** – Earth all metal parts
2. **Overcurrent protection** – Fuses, circuit breakers
3. **Emergency stop** – E-stop pushbuttons
4. **Lockout/Tagout** – Lock power during maintenance

8. Explain the operation of a basic electrical control system with a suitable example.

Answer: Operation involves:

1. **Input/Detection** – Sensor detects condition (e.g., temperature)
2. **Decision/Processing** – Controller compares to setpoint
3. **Action/Output** – Actuator executes required action
4. **Feedback** – Sensor measures result (if closed-loop)

Example – Room Temperature Control:

Thermostat measures temperature → Compare to 25°C

↓

If temp < 25°C → Turn ON heater

If temp > 25°C → Turn OFF heater

↓

Feedback: Sensor continues to monitor

9. Compare open-loop and closed-loop control systems.

Feature	Open-Loop	Closed-Loop
Feedback	No	Yes
Accuracy	Low	High
Error correction	No	Yes
Disturbance handling	Poor	Good
Cost	Low	High
Complexity	Simple	Complex
Example	Toaster	Air conditioner

10. Discuss the importance of electrical control systems in industry.

Answer:

1. **Automation** – Reduces manual labor, increases productivity
2. **Quality control** – Maintains consistent product quality
3. **Safety** – Protects personnel and equipment
4. **Efficiency** – Optimizes energy and resource usage
5. **Monitoring** – Provides real-time data and diagnostics
6. **Flexibility** – Easily reprogrammed for different tasks
7. **Cost reduction** – Reduces waste and downtime

11. Explain the importance of safety in electrical control systems.

Answer:

1. **Personnel protection** – Prevents electric shock and injuries
2. **Equipment protection** – Prevents damage and extends life
3. **Fire prevention** – Protects against electrical fires
4. **Regulatory compliance** – Meets legal requirements
5. **Reliability** – Ensures system operates as intended
6. **Business continuity** – Prevents costly downtime

KEY FORMULAS AND CONCEPTS SUMMARY

Control System Terminology

Term	Definition
Input	Command or reference signal
Setpoint	Desired value of controlled variable
Process	System being controlled
Output	Actual value of controlled variable
Feedback	Information about the output
Error	Difference between setpoint and output
Disturbance	Unwanted influence on the system

Common Components

Component	Symbol Key
Pushbutton (NO)	Normally open - momentary contact
Pushbutton (NC)	Normally closed - momentary contact
Relay coil	Electromagnetic coil
Relay contact	Switch controlled by coil
Timer	Time delay device
Contact	Power switching device